

Morning Paper Report - Tuesday, September 15, 2026

Recent, non-repeated papers matched to visual working memory, visual search, modeling, working-memory representation, neural-network memory models, AI for human memory, and egocentric predictive-coding memory themes.

1. Transcranial Random Aperiodic Stimulation Improves Working Memory Precision

Quirine van Engen, Justin Riddle, Bradley Voytek

2026

Source: Crossref; matched terms: visual working memory, working memory, precision

Link: <https://doi.org/10.64898/2026.09.08.750163>

Goal: Aperiodic neural activity, historically dismissed as noise, is independently modulated from neural oscillations during visual working memory maintenance

Method: In the context of aging, flatter trait aperiodic activity is associated with poorer visual working memory performance Here, we causally test the role of aperiodic neural activity in visual working memory in younger adults

Results: In a randomized, double-blind, placebo-controlled, crossover neurostimulation design (n = 30), we show that online noninvasive tRAS improves visual working memory precision when aperiodic activity is causally steepened

Conclusion: In a randomized, double-blind, placebo-controlled, crossover neurostimulation design (n = 30), we show that online noninvasive tRAS improves visual working memory precision when aperiodic activity is causally steepened Our novel stimulation method paves the way for causal studies of non-oscillatory, aperiodic activity in human cognition, aging, and disease.

2. Object Details Persist in Visual Working Memory Regardless of Category Sufficiency for Search

Ryan Williams, Joseph M. Saito, Keisuke Fukuda, Susanne Ferber

2026

Source: Crossref; matched terms: visual working memory, working memory, attentional guidance

Link: https://doi.org/10.31234/osf.io/cfr7e_v1

Goal: When searching for objects in complex environments, performance often improves with more specific target representations

Method: Across two experiments, observers searched consecutively for images of real-world objects from the same basic-level category

Results: We found that while fewer object-specific details were maintained in VWM when category membership was sufficient, the amount of detail only diminished across repetitions of the exact same object

Conclusion: We therefore examined whether the format of target representations used for search is sensitive to the sustained preservation of object-category membership

3. Individual Differences in Working Memory Capacity and Visual Search While Reading

Ralph S. Redden, Kaylee Eady, Raymond M Klein, Jean Saint-Aubin

2021

Source: Crossref; matched terms: visual search, working memory

Link: <https://doi.org/10.31234/osf.io/pxecn>

Goal: Individual differences in working memory capacity are related to variations in a wide range of cognitive tasks

Method: Participants were presented strings of letters using rapid serial visual presentation and were required to detect all instances of a particular target letter

Results: The hypothesized results from the prose passage were replicated in Experiment 2

Conclusion: Our findings are consistent with recent suggestions that visual search efficiency will be affected by working memory capacity so long as searching is embedded in a context that entails managing resource allocation between concurrent tasks.

4. Memory Constraints in Uncertainty Misestimation: A Computational Model of Working Memory and Environmental Change Detection

Li Xin Lim, Rei Akaishi, Sébastien Hélie

Mathematics, 2025

Source: Crossref; matched terms: working memory, computational model

Link: <https://doi.org/10.3390/math13152431>

Goal: Reinforcement learning models often rely on uncertainty estimation to guide decision-making in dynamic environments

Method: Reinforcement learning models often rely on uncertainty estimation to guide decision-making in dynamic environments

Results: The results show that cognitive load diminished reliance on memory, lowered expected uncertainty, and increased perceptions of environmental volatility

Conclusion: The findings emphasize the importance of memory constraints in uncertainty estimation, highlighting how misestimation of uncertainties is influenced by individual experiences and the capacity of working memory (WM) to store relevant information. These insights contribute to understanding the role of WM in decision-making under uncertainty and provide a framework for exploring the dynamics of reinforcement learning...

5. A Computational Model of Working Memory Based on Spike-Timing-Dependent Plasticity

Qiu-Sheng Huang, Hui Wei

Frontiers in Computational Neuroscience, 2021

Source: Crossref; matched terms: working memory, computational model

Link: <https://doi.org/10.3389/fncom.2021.630999>

Goal: Working memory is closely involved in various cognitive activities, but its neural mechanism is still under exploration

Method: The mainstream view has long been that persistent activity is the neural basis of working memory, but recent experiments have observed that activity-silent memory can also be correctly recalled

Results: We conducted a simulation experiment, and the results are similar to the real experimental results, which suggests that our model is plausible in biology.

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